

2023 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Summary of Question:

Car X (760 kg) at 12.0 m s^{-1} collides with Car Y (840 kg) at 4.0 m s^{-1} . After collision, Car Y moves at 8.5 m s^{-1} .

(a) Velocity of Car X after collision

$$\text{Momentum before} = (760 \times 12.0) + (840 \times 4.0) = 12480$$

$$\text{Momentum after} = (760 \times v_1) + (840 \times 8.5)$$

$$12480 = 760v_1 + 7140 \rightarrow v_1 = 7.03 \text{ m s}^{-1}$$

Answer: 7.03 m s^{-1}

(b) Collision type

$$E_k(\text{before}) = \frac{1}{2}(760 \times 12^2) + \frac{1}{2}(840 \times 4^2) = 61\,440 \text{ J}$$

$$E_k(\text{after}) = \frac{1}{2}(760 \times 7.03^2) + \frac{1}{2}(840 \times 8.5^2) \approx 48\,965 \text{ J}$$

Since KE decreases, collision is inelastic.

Answer: Inelastic

(c) Average force exerted

$$\text{Impulse} = \Delta p = F\Delta t$$

$$\Delta p = 840 \times (8.5 - 4.0) = 3780 \text{ Ns}$$

$$F = 3780 / 0.82 = 4610 \text{ N}$$

Answer: $4.6 \times 10^3 \text{ N}$

(d) Tyre walls

Tyre walls increase the impact time.

Force = $\Delta p / \Delta t$, so increasing Δt reduces the force on the driver.

Answer: Tyre walls reduce impact force

Final Answers

(a) 7.03 m s^{-1}

(b) Inelastic

(c) $4.6 \times 10^3 \text{ N}$

(d) Tyre walls extend collision time, reducing force.

Revision Tips

- Momentum is always conserved in collisions.
- KE conserved only in elastic collisions.
- Impulse = change in momentum.
- Safety features (airbags, crumple zones, tyre walls) increase impact time.